

Module 11

Supporting Network Security Design

Learning Objectives

- Explain the importance of network segmentation and use of trusted and untrusted zones.
- Describe security implications of internet of things (IoT) and industrial internet of things (IIoT).
- Explain the importance of physical security.

Lesson 11.1

Zone-Based Security

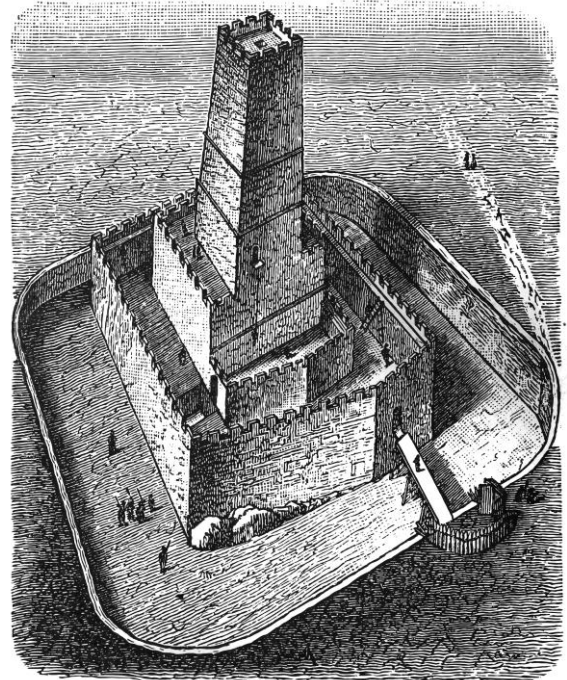
Zone-Based Security

Definition

- Network security model
- Organizes network into security zones
- Zones represent a region with specific security requirements

Security Model

- Ensures that traffic flows securely between zones
- Controls access
- Minimizes risk





Set Network Location

Select a location for the 'Network 12' network

This computer is connected to a network. Windows will automatically apply the correct network settings based on the network's location.



Home network

If all the computers on this network are at your home, and you recognize them, this is a trusted home network. Don't choose this for public places such as coffee shops or airports.



Work network

If all the computers on this network are at your workplace, and you recognize them, this is a trusted work network. Don't choose this for public places such as coffee shops or airports.



Public network

If you don't recognize all the computers on the network (for example, you're in a coffee shop or airport, or you have mobile broadband), this is a public network and is not trusted.



Treat all future networks that I connect to as public, and don't ask me again.

[Help me choose](#)

Network Security Zones

Private Security Networks

Contain sensitive resources like internal servers, databases, and critical systems

Access is tightly controlled

Private Client Networks

Reserved for internal users and devices

Includes workstations, laptops, and other endpoints

Guest Network

For external visitors or temporary access

Limited privileges and isolated from critical resources

Public Server Network

Houses publicly accessible servers (e.g., web servers)

Separated from internal networks to prevent direct attacks

Perimeter Networks

Definition

- A secure buffer zone between an organization's internal network and the external, untrusted Internet

Characteristics

- Controls and filters traffic between internal and external networks
- Houses publicly accessible servers (e.g., web, email)
- Security appliances (firewalls, intrusion detection systems) are deployed here
- Separates sensitive data and systems from the public-facing network
- Provides policy enforcement, inspection, and auditing

Location

- Between external (untrusted) networks (e.g., Internet) and internal networks

Screened Subnet

A firewall architecture that defines three separate subnets:

External Router

Separates the external network from the perimeter network

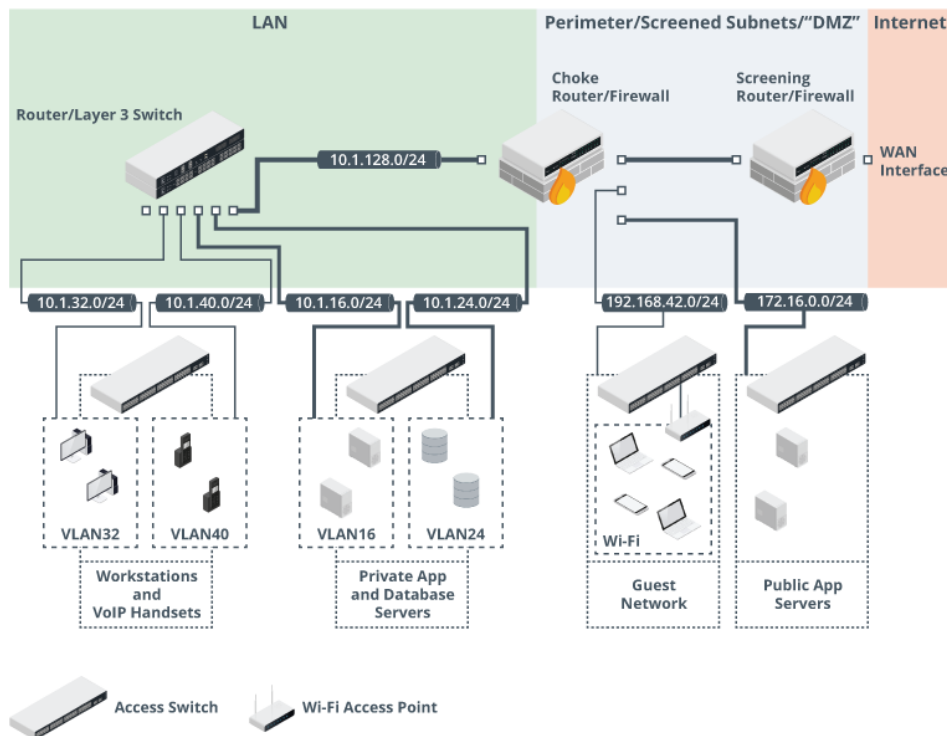
Internal Router

Separates the perimeter network from the internal network

Perimeter Network

Hosts servers accessible from both internal and external networks

Screened Subnet Example



Intrusion Detection and Prevention

IDS (Intrusion Detection System)

- Monitors network traffic, detects known attack patterns, and alerts administrators
- Traffic keeps flowing even during an attack
- Useful for quick setup but may delay response time

IPS (Intrusion Prevention System)

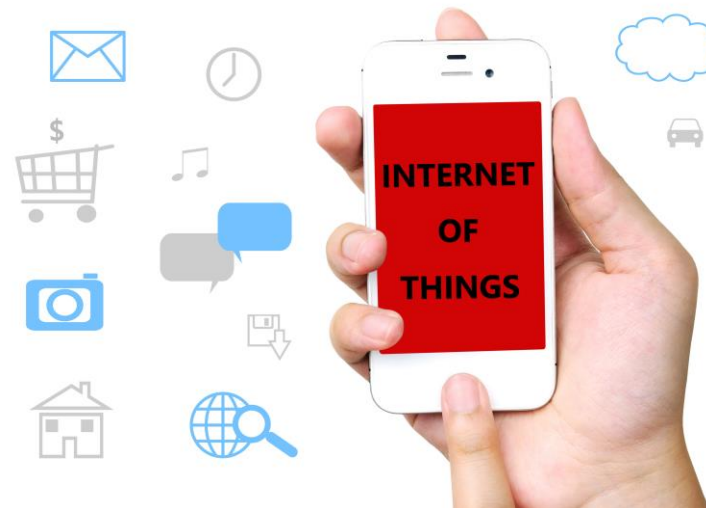
- Stops attacks immediately upon detection
- Traffic halts until investigation and decision
- Prioritizes prevention over alerting
- Ideal for critical assets but can impact legitimate traffic

Lesson 11.2

Internet of Things

Think About It: IoT

What is the internet of things (IoT)?



IoT Devices

Consumer devices

Designed for personal use and convenience

Often controlled through mobile devices

Industrial devices

Monitor and control various aspects of industrial operations

Improve efficiency and safety

Healthcare devices

Support healthcare providers

Help provide personalized, timely patient care

Agricultural devices

Enhance precision farming

Provide real-time data to optimize crop yield and livestock health

Smart city devices

Help manage public resources efficiently

Improve the quality of urban life

Industrial Embedded Systems in IoT

Definition

- Specialized computing systems implanted in larger devices/systems
- Performs dedicated functions/tasks within industrial applications

Industrial Control Systems (ICS)

- Designed to manage and control industrial processes
- Used for automating plant operations

Supervisory Control and Data Acquisition (SCADA)

- Focuses on real-time data acquisition and control
- Monitors and controls equipment and conditions

IoT Networking

IoT Network Integration

- Use unique identifiers for Internet connectivity, sometimes via intermediaries
- Standard networking products (Ethernet, Wi-Fi) might not always suit IoT device needs

IoT Networking Standards

Operational Technology (OT) Networks

- Industrial settings, optimized for real-time data transfer
- Uses serial protocols or industrial Ethernet for deterministic transfers

Cellular Networks for IoT

- Narrowband-IoT (NB-IoT)
 - Low-power LTE variant, suitable for sending small data packets in challenging environments
- LTE Machine Type Communication (LTE-M)
 - Supports higher bandwidth (up to 1 Mbps) for more data-intensive IoT applications

IoT Network Security

Definition & Importance

Protecting the network connecting IoT devices to back-end systems on the internet

Main Challenges

Increase the attack surface

Devices with weak security

Mitigation Strategies

Segmentation

Regular audits

Employee Training

Managing IoT Security Risks

Consumer-Grade Devices

Risk: Vulnerabilities (weak default settings/inadequate patch management)

Management: Secure and patch IoT administrative interfaces

Smart Buildings

Risk: Compromise of entry mechanisms and climate/lighting control components

Approach: Configuration management and isolated network segments

ICS/SCADA

Challenges: Vulnerability (linked OT/IT networks) exploited by threat actors

Solution: Isolate management traffic and monitor cross-network links (access controls)

Lesson 11.3

Physical Security

Physical Security

Why Physical Security is Important

- First line of defense in protecting network resources
- Can lead to data breaches, hardware theft, and vandalism

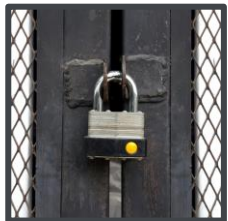
Risks of Inadequate Physical Security

- Data Theft
- Hardware Damage
- Unauthorized Access

Mitigation Strategies

- Access Control Systems
- Surveillance Systems
- Environmental Controls
- Regular Audits

Locks and Secure Entry Systems



Locks and Secure Entry Systems

- Physical barriers and locks against unauthorized access

Locks and Badge Readers

- **Electronic Locks:** Secures entries electronically
- **Badge Reader:** Identifies and authorizes access via smart badges
- **Biometric Locks:** Authenticates using physical features



Rack System Locks

- **Secure Cabinets and Enclosures:** Protects IT assets from unauthorized access
- **Colocation Cages:** Provides separate access in shared data centers



Camera-Based Surveillance Systems

Detection-Based Controls

- Act as a crucial layer of defense
- Ensure resilient security perimeters

Surveillance Applications

- Monitoring both perimeter areas and internal security zones
- Manned by security personnel or automated through video surveillance



Camera Types

Fixed Cameras

- Recording all individuals passing through specific entry points
- Examples: access control vestibules

PTZ (Pan-Tilt-Zoom) Cameras

- Versatility for extensive area coverage
- Ability to zoom in on details and track movements within large spaces

CCTV Network Integration

- Cameras typically connect via coaxial cable to a multiplexer
- Enable image display across multiple screens
- Include both control and recording functionalities
- Integration into IP networks for enhanced flexibility and efficiency
- Smaller IP cameras support Power over Ethernet (PoE)

Geofencing Overview

Geolocation and its Uses

- Identifies the physical position of a device using network attributes
- Location services for determining a device's current position

Location Services

- GPS (Global Positioning System)
- IPS (Indoor Positioning System)

Geofencing Explained

- A virtual boundary created in real-world geography
- Controlling device functionalities (camera and video)
- Context-aware authentication

Geofencing Overview

Applications in Organizations

- Creation of virtual perimeters around specific areas (e.g., office properties)
- Restricts device functions (e.g., camera use outside the designated area)

How Does it Work?

- Device position identified through location services
- Enforcement and policy management facilitated by Mobile Device Management (MDM) software



Summary

- **Perimeter Defense:** Utilize firewalls/proxy servers in screened subnets
- **Internal Zone Protection:** Implement firewalls for internal zones/hosts
- **Intrusion Detection/Prevention:** Deploy IDS/IPS behind firewalls
- **IoT Management:** Develop a strategy for segmented network zones
- **Site Access Control:** Assess and implement access control measures